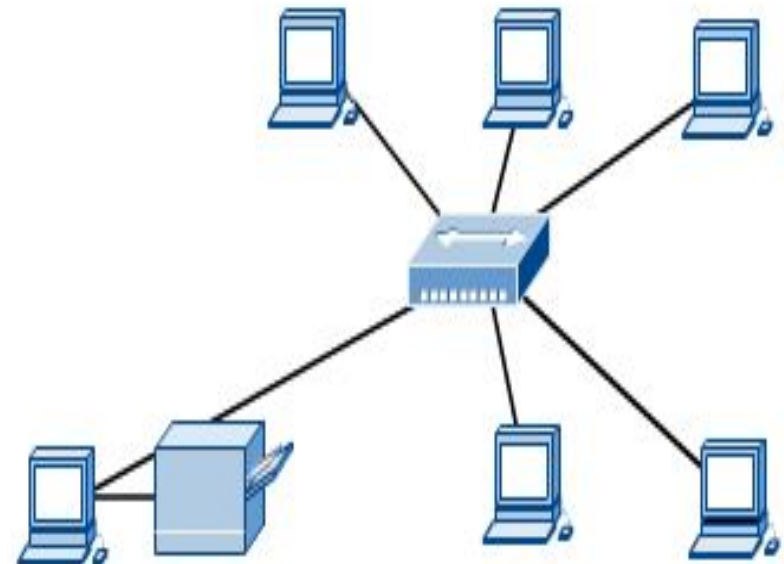




INTRODUCTION TO COMPUTER NETWORKS

Computer Networks

- **Computer network connects two or more autonomous computers.**



- **The computers can be geographically located anywhere.**

LAN, MAN & WAN

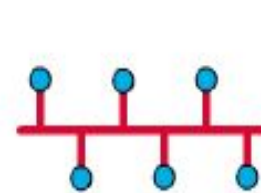
- Network in small geographical Area (Room, Building or a Campus) is called LAN (Local Area Network)
- Network in a City is call MAN (Metropolitan Area Network)
- Network spread geographically (Country or across Globe) is called WAN (Wide Area Network)

Applications of Networks

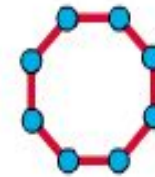
- **Resource Sharing**
 - Hardware (computing resources, disks, printers)
 - Software (application software)
- **Information Sharing**
 - Easy accessibility from anywhere (files, databases)
 - Search Capability (WWW)
- **Communication**
 - Email
 - Message broadcast
- **Remote computing**
- **Distributed processing (GRID Computing)**

Network Topology

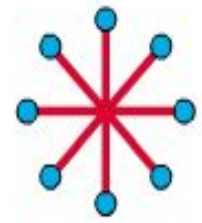
- The network topology defines the way in which computers, printers, and other devices are connected. A network topology describes the layout of the wire and devices as well as the paths used by data transmissions.



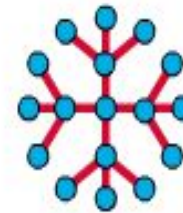
Bus Topology



Ring Topology



Star Topology



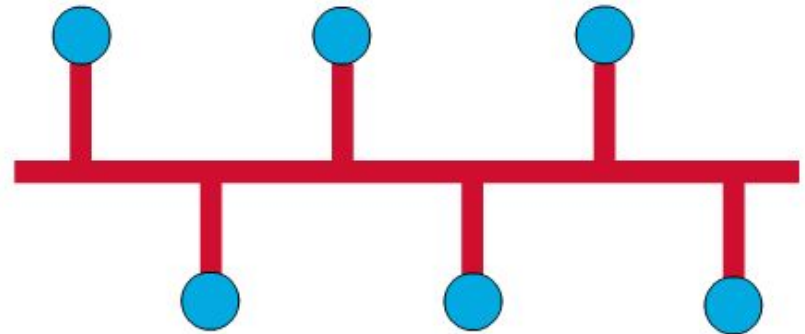
Extended Star
Topology



Mesh
Topology

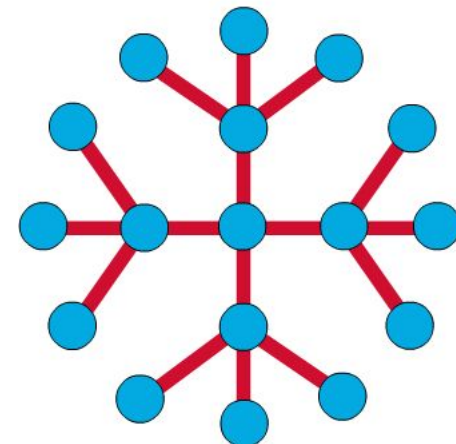
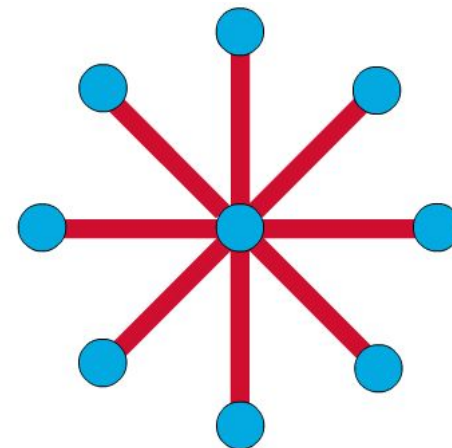
Bus Topology

- Commonly referred to as a linear bus, all the devices on a bus topology are connected by one single cable.



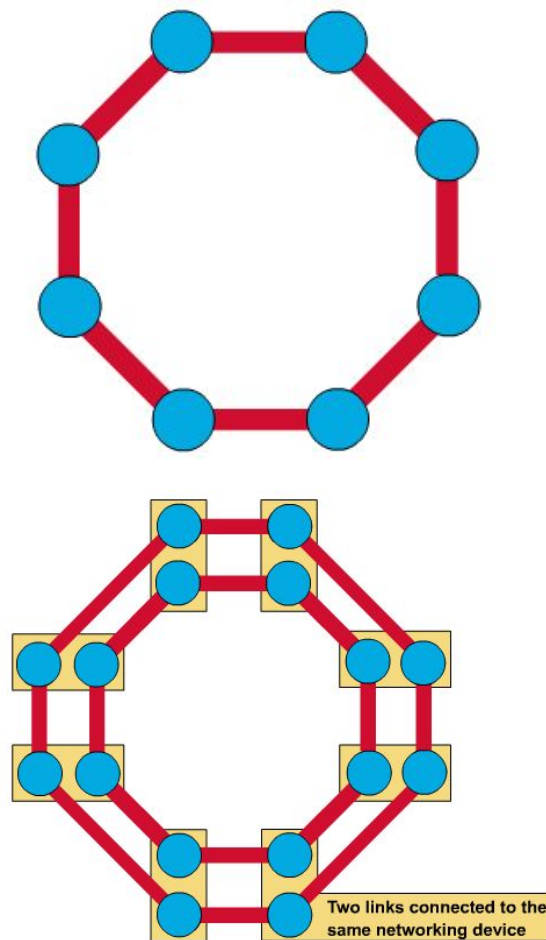
Star & Tree Topology

- The star topology is the most commonly used architecture in Ethernet LANs.
- When installed, the star topology resembles spokes in a bicycle wheel.
- Larger networks use the extended star topology also called tree topology. When used with network devices that filter frames or packets, like bridges, switches, and routers, this topology significantly reduces the traffic on the wires by sending packets only to the wires of the destination host.



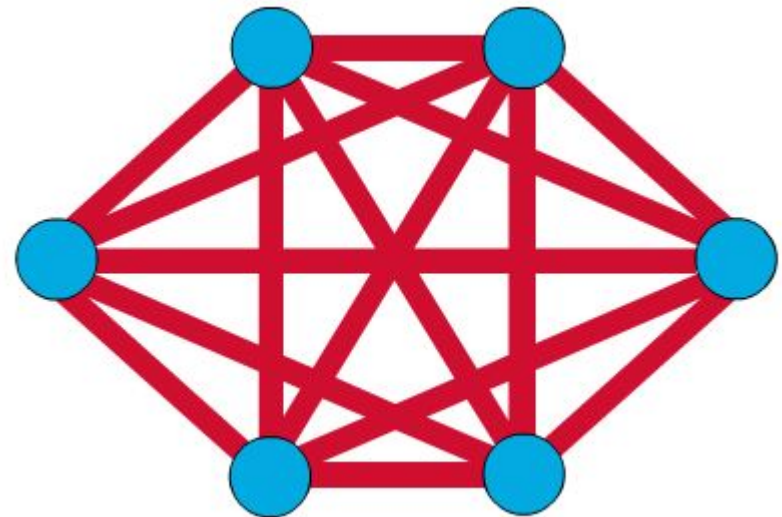
Ring Topology

- A frame travels around the ring, stopping at each node. If a node wants to transmit data, it adds the data as well as the destination address to the frame.
- The frame then continues around the ring until it finds the destination node, which takes the data out of the frame.
 - Single ring – All the devices on the network share a single cable
 - Dual ring – The dual ring topology allows data to be sent in both directions.



Mesh Topology

- The mesh topology connects all devices (nodes) to each other for redundancy and fault tolerance.
- It is used in WANs to interconnect LANs and for mission critical networks like those used by banks and financial institutions.
- Implementing the mesh topology is expensive and difficult.

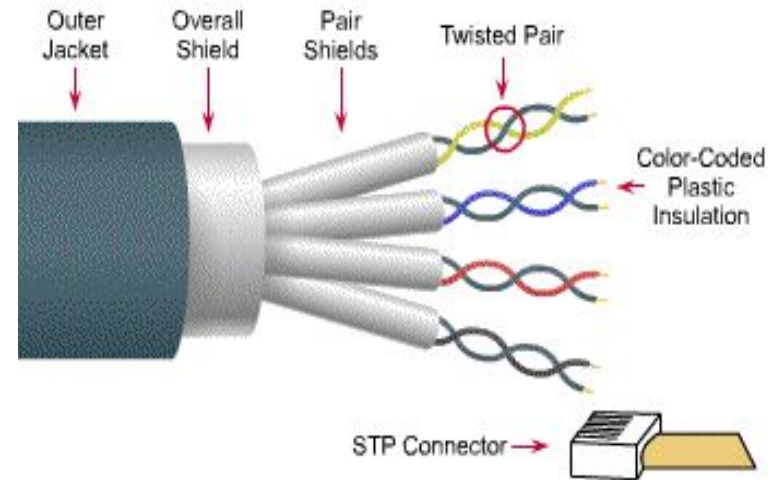


Network Components

- **Physical Media**
- **Interconnecting Devices**
- **Computers**
- **Networking Software**
- **Applications**

Networking Media

- Networking media can be defined simply as the means by which signals (data) are sent from one computer to another (either by cable or wireless means).



- Speed and throughput: 10-100 Mbps
- Cost per node: Moderately expensive
- Media and connector size: Medium to Large
- Maximum cable length: 100m (short)

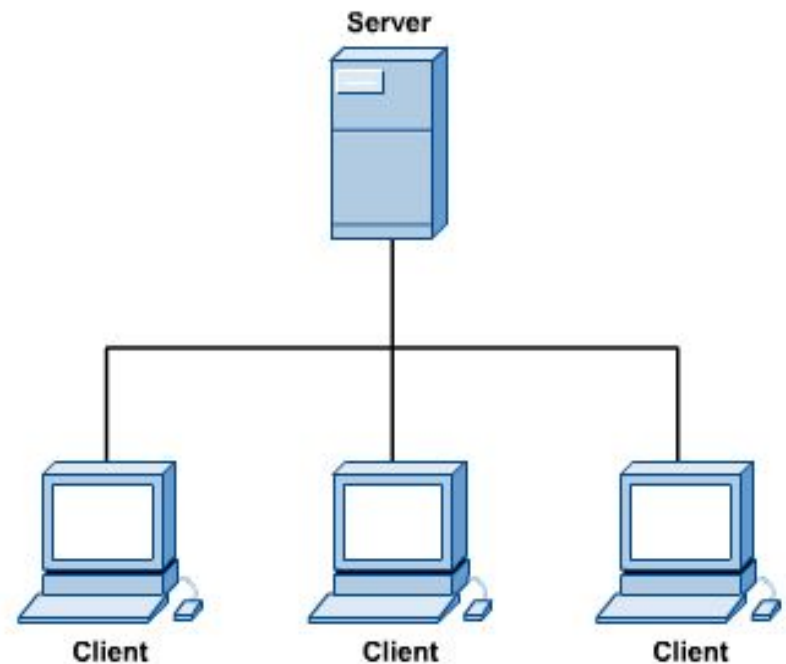
Networking Devices

- HUB, Switches, Routers, Wireless Access Points, Modems etc.

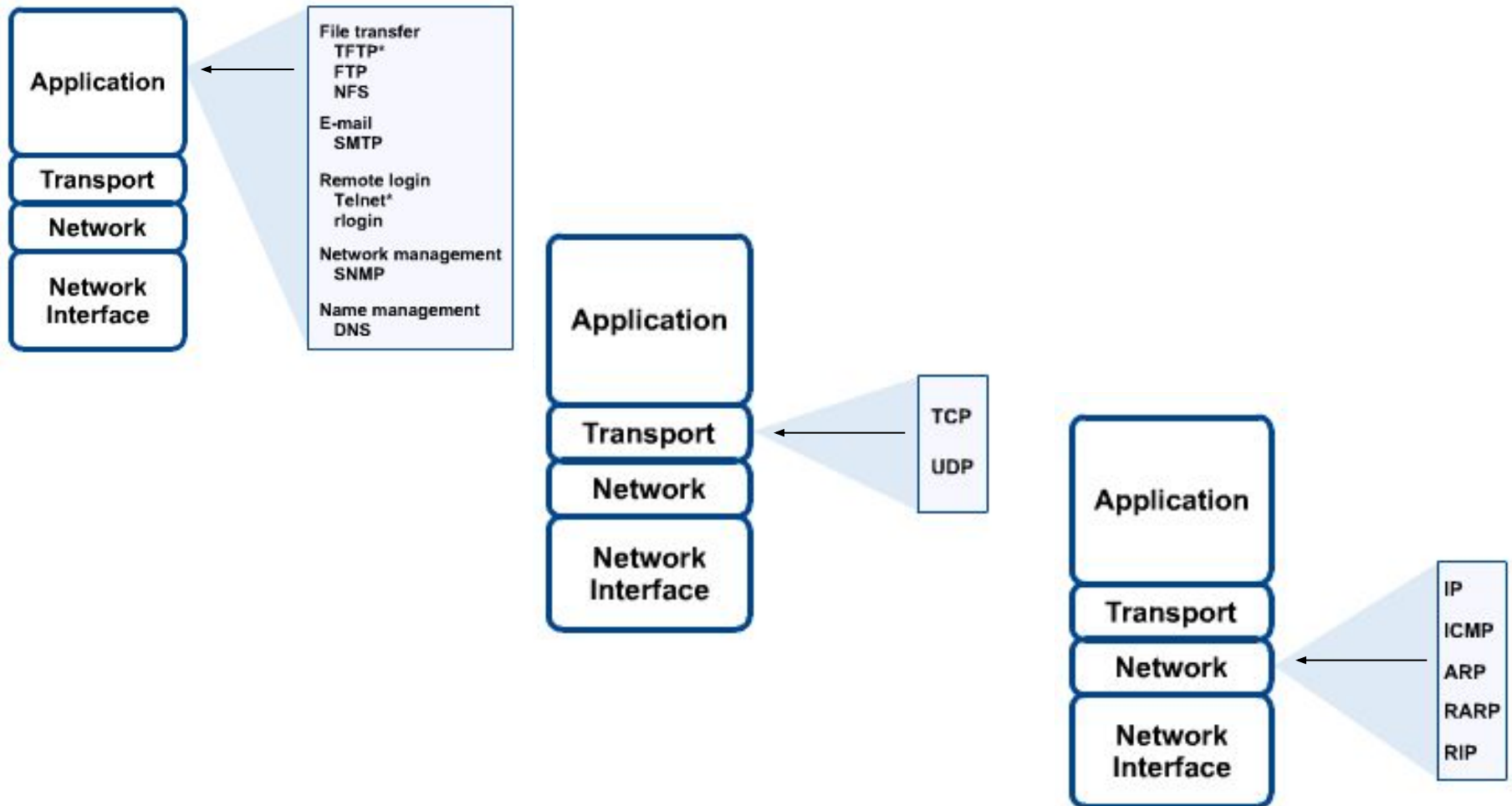


Computers: Clients and Servers

- In a client/server network arrangement, network services are located in a dedicated computer whose only function is to respond to the requests of clients.
- The server contains the file, print, application, security, and other services in a central computer that is continuously available to respond to client requests.



Networking Protocol: TCP/IP



Applications

- E-mail
- Searchable Data (Web Sites)
- E-Commerce
- News Groups
- Internet Telephony (VoIP)
- Video Conferencing
- Chat Groups
- Instant Messengers
- Internet Radio

